

Program 8 – Execution Steps

Step 1: Create project directory

```
mkdir Program8  
cd Program8
```

Step 2: Create server.js

```
nano server.js
```



```
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible: ~/Desktop/devopslab/Program-8  
GNU nano 7.2 server.js  
const http = require("http");  
const port = 3000;  
  
const server = http.createServer((req, res) => {  
  res.end("Hello from Node.js running inside Kubernetes!");  
});  
  
server.listen(port, () => {  
  console.log(`Server running at port ${port}`);  
});
```

Step 3: Create ConfigMap YAML

```
nano nodejs-configmap.yml
```



```
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible: ~/Desktop/devopslab/Program-8  
GNU nano 7.2 nodejs-configmap.yml  
apiVersion: v1  
kind: ConfigMap  
metadata:  
  name: nodejs-app-config  
data:  
  server.js: |  
    const http = require("http");  
    const port = 3000;  
  
    const server = http.createServer((req, res) => {  
      res.end("Hello from Node.js running inside Kubernetes!");  
    });  
  
    server.listen(port, () => {  
      console.log(`Server running at port ${port}`);  
    });
```

Step 4: Create Pod YAML

```
nano nodejs-pod.yml
```

```
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible: ~/Desktop/devopslab/Program-8
GNU nano 7.2 nodejs-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nodejs-pod
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs
      image: node:18
      command: ["node", "/usr/src/app/server.js"]

      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app

      ports:
        - containerPort: 3000

  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config
```

Step 5: Apply the ConfigMap and Pod

kubectl apply -f nodejs-configmap.yaml

kubectl apply -f nodejs-pod.yaml

Check pod status

kubectl get pods

```
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/Program-8$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
backup-job-29468485-mscd5           0/1     Completed 0           13m
backup-job-29468490-24p46           0/1     Completed 0           9m31s
backup-job-29468495-v64n2           0/1     Completed 0           4m31s
my-cron-29468485-vp5wg               0/1     Completed 0           13m
my-cron-29468490-zwtmm               0/1     Completed 0           9m31s
my-cron-29468495-76l2f               0/1     Completed 0           4m31s
nginx-deployment-6f9664446b-n6d2r   1/1     Running   0           10m
nginx-deployment-6f9664446b-z2vs6   1/1     Running   0           10m
nodejs-pod                           1/1     Running   6 (13m ago) 31d
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/Program-8$
```

Step 8: Access the Node.js application

kubectl port-forward pod/nodejs-pod 3000:3000

Step 9: Open browser

<http://localhost:3000>

